

MATH DTP REPORT

Problem Statement:

How might we help solar plant managers in India accurately forecast short-term (15-minute interval) solar energy output using real-time environmental data, while also detecting potential anomalies in plant performance?

Dataset Description

We utilize the **Plant 1 Generation Data** from the Kaggle dataset titled “**Solar Power Generation Data**”, which contains inverter-level power generation readings from a solar power plant located in India. The data is recorded at 15-minute intervals over a span of 34 days.

In addition, we incorporate the **Plant 1 Weather Sensor Data**, provided as a separate file within the same dataset. These two datasets will be merged later to construct the complete training dataset for our analysis. This dataset contains about 3183 field entries.

Column Description for each file in the dataset are as follows:

Plant_1_Generation_Data.csv

Column Name	Description
DATE_TIME	Timestamp of the power generation reading, recorded at 15-minute intervals.
PLANT_ID	Unique identifier for the solar power plant
SOURCE_KEY	Unique identifier for each inverter within the plant.
DC_POWER	Direct Current (DC) power output from the inverter, measured in Watts.
AC_POWER	Alternating Current (AC) power output after DC-AC conversion, measured in Watts.
DAILY_YIELD	Energy generated by the inverter during the current day, in Watt-hours (Wh). Resets daily.
TOTAL_YIELD	Cumulative energy generated by the inverter since installation, in Watt-hours (Wh).

Plant_1_Weather_Sensor_Data.csv

Column Name	Description
DATE_TIME	Timestamp of the weather sensor reading, recorded at 15-minute intervals.
PLANT_ID	Identifier for the solar power plant (same as above).
SOURCE_KEY	Identifier for the weather sensor.
AMBIENT_TEMPERATURE	Ambient (surrounding) air temperature in degrees Celsius (°C).
MODULE_TEMPERATURE	Surface temperature of the solar panel/module in degrees Celsius (°C).
IRRADIATION	Solar irradiance measured in Watts per square meter (W/m ²).

Clean and Analyse the Dataset

1. **Convert DATE_TIME columns in both datasets to a consistent datetime format.** This allows us to accurately merge the datasets by grouping them via the time stamps.

- Each timestamp has 21 rows in the generational data (one for each inverter)
- We sum up the power and yield values across all 21 inverters at each timestamp giving us one row per timestamp, accurately reflecting the plant's total output.
- Merge aggregated generation data with weather data grouping by DATE_TIME

2. Analysing the data by plotting the **histogram** for one of the most promising features “**IRRADIATION**”.

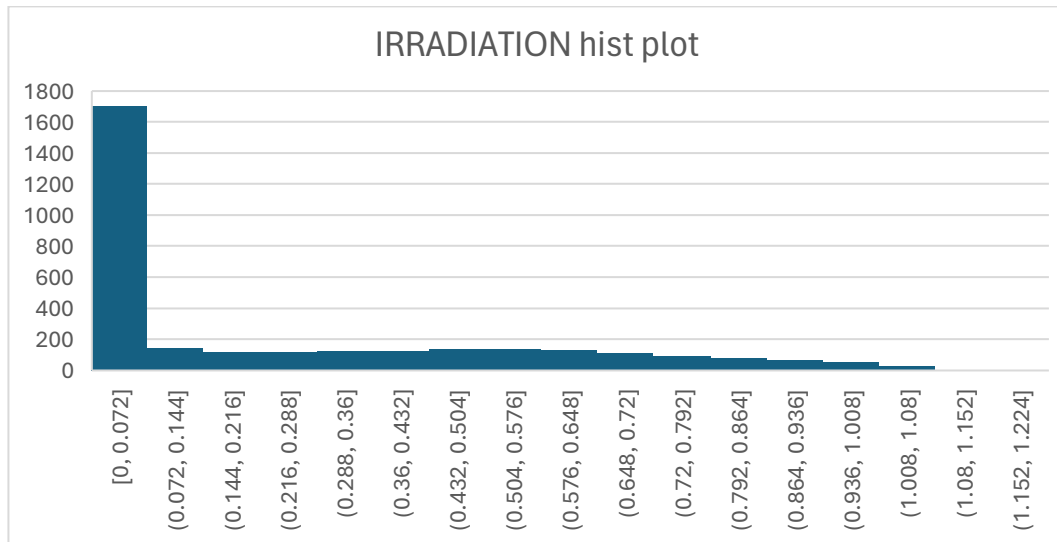


Figure 1: IRRADIATION Histogram Plot

Plot demonstrates tail heavy characteristics which means the number of values which are close to 0 are high. This might bias our dataset later. We'll try to mitigate this problem after further analysis.

3. Plotting the **IRRADIATION vs DATE TIME** to get a feel for the Irradiation data pattern:

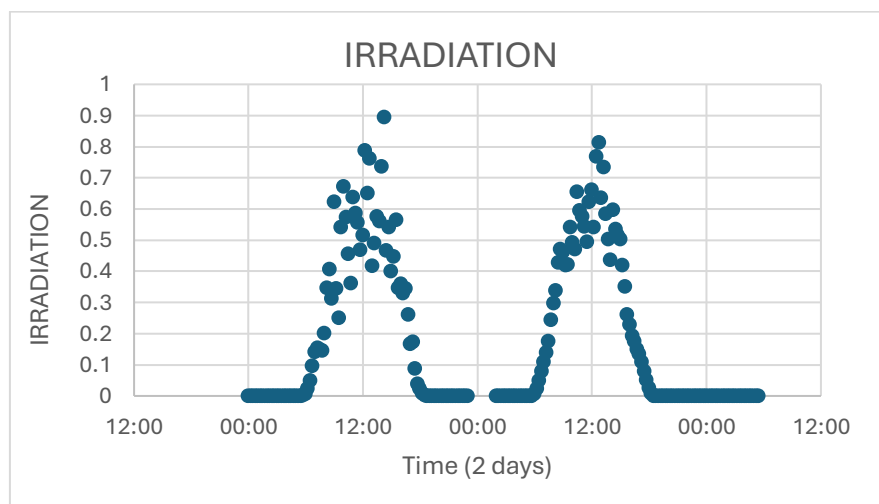


Figure 22: Irradiation over time (2days)

The plot exhibits a symmetrical distribution centered around midday, which is expected as it aligns with typical daily solar patterns. This explains the tail-heavy nature of the previous histogram since almost half the Irradiation values will be 0 skewing the data to the left. We can mitigate this problem by removing the nighttime data from the training set.

4. Drop unused and redundant columns like PLANT_ID_X and PLANT_ID_Y as they are constant values, SOURCE_KEY as it is not useful for modelling, DAILY_YIELD as we are using the new sum of AC_POWER and DC_POWER instead.

5. We observed that **AC Power** has a **perfect linear relationship** with **DC Power** ($R^2 = 1$), as expected due to the physical process of DC-to-AC conversion. Therefore, we selected **AC Power** as the **target variable** since it represents the final output of the plant.

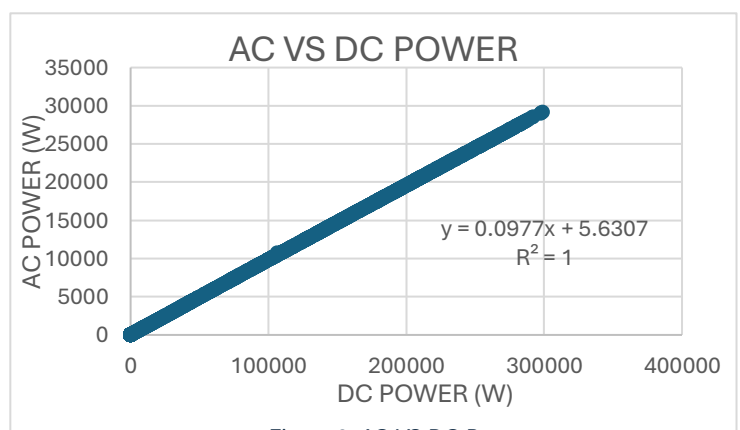


Figure 3: AC VS DC Power

5. Visualizing key features:

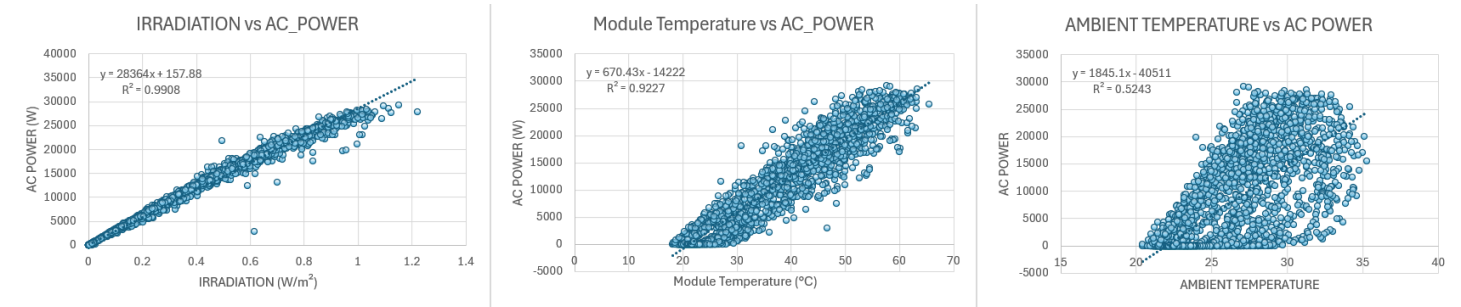


Figure 4: Import features vs AC Power

- **Irradiation vs AC Power** shows a **strong linear correlation**, making it a key predictor.
- **Module Temperature vs AC Power** also demonstrates a **very strong relationship** ($R^2 = 0.9227$), supporting its inclusion.
- **Ambient Temperature vs AC Power** shows a **weaker correlation** ($R^2 = 0.5243$), likely affected by **nighttime skew**. With proper cleaning, it could still contribute meaningfully.

6. Feature Engineering: we first engineered features that capture both environmental and temporal influences on solar power output. We introduced two cyclical time-based variables.

$$SIN_TIME = \sin\left(\frac{\text{hour} + \frac{\text{minute}}{60}}{24} \cdot 2\pi\right); \quad COS_TIME = \cos\left(\frac{\text{hour} + \frac{\text{minute}}{60}}{24} \cdot 2\pi\right)$$

This allows the model to better recognize periodic fluctuations in solar production throughout the day.

7. Final features:

Type	Name	Description
Feature (X)	IRRADIATION	Incoming solar energy, the most direct driver of AC power.
Feature (X)	MODULE_TEMPERATURE	Temperature of the solar module, impacted by irradiation and influencing efficiency.
Feature (X)	AMBIENT_TEMPERATURE	Surrounding air temperature, affecting module performance.
Feature (X)	SIN_TIME	Sine of the time of day, encoding cyclic solar generation patterns.
Feature (X)	COS_TIME	Cosine of the time of day, encoding cyclic solar generation patterns.
Target (y)	AC_POWER	The alternating current power output from the plant (in watts).

8. To ensure model stability, we applied **Z-score normalization** to each input feature. This standardization ensures that features contribute equally to the optimization process by centring their distributions around zero with unit variance.

9. After normalization, we constructed the design matrix by **appending a column of ones to represent the intercept term**, preparing the data for linear regression modelling

Building Model

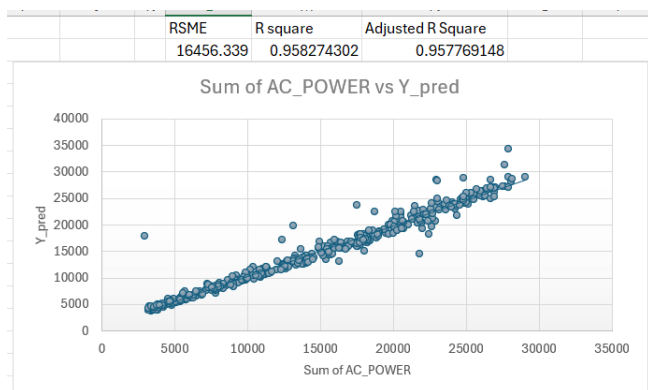
We trained a multiple linear regression model to predict AC power using the **closed-form solution** (Normal Equation), not gradient descent. All input features were first normalized for consistent scaling. A design matrix X was constructed with an added bias term (intercept), and the optimal model parameters β were computed analytically using: $\beta = (X^T X)^{-1} X^T y$.

This formula minimizes the **Mean Squared Error (MSE)** between predicted and actual AC power values, providing an exact solution without iterative optimization.

Evaluating the Model

To evaluate model performance, we used the Root Mean Squared Error (RMSE), which measures the average magnitude of prediction errors. Lower RMSE values indicate better predictive accuracy. The RMSE is defined as:

$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$ where (y_i) are the actual values and $(\hat{y}_i)$ are the predicted values. We also computed the coefficient of determination (R^2), which quantifies the proportion of variance in the target variable explained by the model. It is given by: $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$ where $(\bar{y})$ is the mean of the actual values. To account for the number of features used, we also calculated the adjusted (R^2), which penalizes model complexity: $Adjusted\ R^2 = 1 - (1 - R^2) \cdot \frac{n-1}{n-p-1}$ where (n) is the number of observations and (p) is the number of features. Our model achieved an RMSE of 16456.339 and an (R^2) score of 0.9583, indicating that it explains approximately 95% of the variance in the target variable. The adjusted R^2 is 0.9577.

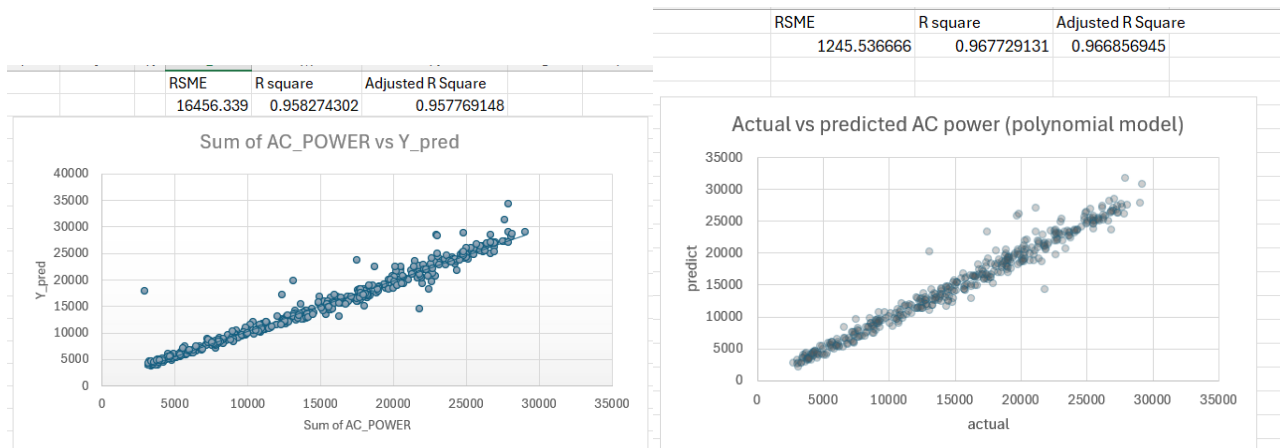


Improving the Model (Polynomial Regression)

To improve upon the baseline linear regression model, we implemented a polynomial regression approach. Using Excel, we generated higher-order terms for the input features and tested different polynomial degrees through a combination of grid search and trial-and-error. After evaluation, a **degree of 3** was selected as it struck a good balance between capturing the non-linear relationships in the data and avoiding underfitting. The updated model was then used to generate predictions, and a new graph was produced to compare its performance directly against the original linear regression model.

Evaluation of the new model

Comparison with the old model: The improved model's residuals were more symmetrically distributed and had smaller magnitude, and the actual vs predicted scatter aligned closely along the ideal line, confirming enhanced predictive performance. The results are shown in the graphs below.



Conclusion.

Polynomial regression, coupled with careful cross-validation, improved prediction accuracy and captured nonlinear relationships between environmental factors and power generation. To further refine the model, additional feature engineering (e.g., interaction terms or time-of-day segments), more sophisticated models (e.g., tree-based ensembles), and robust anomaly detection could be explored.

Name of students and ID:

1. Somesh Agrawal (1009681): Contributed to the selection and visualization of the dataset.
2. Can Viet Minh: Assisted in preprocessing the data and developing the regression models.
3. Kang: Worked on evaluating model performance and refining the model to improve accuracy.
4. Erwin: Handled data cleaning and implemented the code for residual analysis.
5. Xavier: Designed the presentation layout and contributed to writing the final report.